When Does Temporal Memory Help? Selective SSM vs. MLP in RL-Based Multi-Target DOA Track Management

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***Abstract*—We address when temporal memory helps in reinforcement learning (RL)-based multi-target DOA track management. We prove that a selective SSM encoder outperforms a memoryless MLP if and only if the observation is *not* a sufficient statistic for the policy-relevant history—a condition satisfied by raw high-dimensional COP spectra over long episodes but not by hand-crafted low-dimensional features. We instantiate this as Mamba-COP-RL: a Mamba SSM encoder coupled with PPO, operating on the 181-dim COP spatial spectrum. On CombatTrackEnv (four combat scenarios), Mamba-COP-RL improves GOSPA by up to 25.2% over fixed GM-PHD thresholds and 6.7% over MLP-PPO, with gains ordered by temporal dependency horizon as predicted. The 41.4 KB INT8 model runs at 3.4 ms/scan on STM32H7 Cortex-M7.**

***Index Terms*—direction of arrival, multi-target tracking, reinforcement learning, selective state space model, Mamba, GM-PHD, edge AI, CombatTrackEnv, sufficient statistic**

I. Introduction

Multi-target DOA tracking with compact arrays is a core challenge in defense, speech enhancement, and wireless communications. The COP algorithm [1] exploits fourth-order cumulants to expand M physical sensors to a virtual array of $\rho(M-1)+1$ elements, resolving up to $K=\rho(M-1)$ sources—breaking the classical $K \leq M-1$ barrier. Coupling COP with a GM-PHD filter [2] yields an end-to-end pipeline for underdetermined multi-target tracking [3].

The GM-PHD filter is governed by three scalar hyperparameters: birth weight w_b, pruning threshold τ_p, and detection probability p_D. Fixed values degrade sharply in non-stationary combat environments: jamming collapses SNR, stealth targets vanish and reappear, saturation attacks flood the scene, and formations fragment. RL offers a data-driven solution—map filter state to parameters adaptively.

Prior RL-based filter tuning uses memoryless MLP policies [4], [5]. A natural extension is to add temporal memory (LSTM, GRU, SSM) to the policy. However, recurrent policies add complexity and can *hurt* performance if temporal information is already saturated by hand-crafted summary statistics.

Central question: *when does temporal memory help?*

Our answer (Proposition 1): SSM outperforms MLP when the observation is *high-dimensional raw signal* and the episode is *long* enough to create multi-scan causal dependencies. We verify this with a controlled ablation:

- **12-dim** hand-crafted statistics, 40 scans $\Rightarrow$ MLP wins; recurrent policies hurt.
- **183-dim** raw COP spectrum, 200 scans $\Rightarrow$ Mamba wins on Stealth and Jamming.

Contributions:

1) A proposition with complete proof characterizing when SSM temporal memory improves RL-based track management.
2) **Mamba-COP-RL**: selective SSM encoder + PPO actor-critic; 42 K params, 41.4 KB INT8.
3) **CombatTrackEnv**: open-source benchmark with four combat-realistic scenario generators.
4) Comprehensive ablation including GRU-PPO, LSTM-PPO, d_state sensitivity, and episode-length dependency.
5) Edge deployment analysis: 3.4 ms/scan on STM32H7 Cortex-M7.

II. Related Work

Prior RL-based GM-PHD tuning uses memoryless MLP policies with low-dimensional hand-crafted features [4], [5]; Ewers *et al.* [6] compared BiGRU and self-attention against MLP for radar tracking RL but did not formally characterize when recurrence helps. Hausknecht and Stone [7] showed DRQN improves performance in POMDPs; Pleines *et al.* [8] analyzed generalization limits of recurrent PPO. Mamba [9] extends S4 [10] with input-dependent selectivity, enabling adaptive gating of irrelevant scans (e.g., zero-signal stealth windows) while retaining trajectory history in $\mathbf{h}_t$—a capability absent from fixed-rate GRU/LSTM. Ota [11] applied Mamba to offline RL; Han *et al.* [12] to audio signal processing. We are the first to deploy Mamba as an *online* RL encoder for adaptive DOA filter tuning and to formally characterize when temporal memory helps.

III. Background

A. COP-DOA and GM-PHD Tracking

The COP pseudo-spectrum is:

$$P_{\mathrm{COP}}(\theta) = \frac{1}{\|\mathbf{P}_{\mathbf{A}}^{\perp}\mathbf{a}_v(\theta)\|^2} \tag{1}$$

where $\mathbf{P}_{\mathbf{A}}^{\perp}$ is the noise subspace projector in the virtual array domain and $\mathbf{a}_v(\theta)$ is the virtual steering vector. Evaluated at $N_\theta=181$ angles, P_{COP} is the primary observation for our RL policy.

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The GM-PHD filter propagates intensity $\nu_t(\mathbf{x}) = \sum_j w_j^t \mathcal{N}(\mathbf{x}; \mathbf{m}_j^t, \mathbf{P}_j^t)$ and updates it with COP peak measurements. Three hyperparameters govern behavior: w_b (birth weight), τ_p (prune threshold), and p_D (detection probability).

B. Selective State Space Models

A selective SSM [9] defines:

$$\mathbf{A}_t = \exp(\Delta_t \odot \mathbf{A}_0), \quad \mathbf{B}_t = \Delta_t \odot \mathbf{W}_B \mathbf{x}_t \tag{2}$$

$$\mathbf{h}_t = \mathbf{A}_t \mathbf{h}_{t-1} + \mathbf{B}_t \mathbf{x}_t, \quad y_t = \mathbf{c}_t^\top \mathbf{h}_t \tag{3}$$

where $\Delta_t, \mathbf{B}_t, \mathbf{c}_t$ are input-affine functions. Selectivity allows the model to learn *which* inputs to retain in $\mathbf{h}_t$ and which to discard, enabling long-range compression without fixed forgetting rates—a key advantage over GRU and LSTM.

IV. Mamba-COP-RL

A. When Does Temporal Memory Help?

Proposition 1 (SSM Advantage Condition)**.** *Let $f : \mathbb{R}^{d_o} \to \mathbb{R}^{d_f}$ be any deterministic observation summary ($d_f \le d_o$). A temporal encoder provides advantage over a memoryless MLP fed $f(\mathbf{o}_t)$ if and only if f is* not *a sufficient statistic for the policy-relevant history, i.e.,*

$$\mathcal{I}(\mathbf{o}_t;\, \mathbf{o}_{1:t-1}) \;>\; \mathcal{I}(f(\mathbf{o}_t);\, \mathbf{o}_{1:t-1}) \tag{4}$$

where $\mathcal{I}(\cdot;\cdot)$ denotes mutual information.

Proof sketch. (*Necessity.*) If equality holds, f is a sufficient statistic by the data processing inequality [13], so the MLP on $f(\mathbf{o}_t)$ already accesses all history-relevant information and no temporal encoder can improve it. (*Sufficiency.*) If strict inequality holds, there exists $g(\mathbf{o}_t, \mathbf{h}_{t-1})$ with strictly higher mutual information with $\mathbf{o}_{1:t-1}$. Since a sufficiently expressive SSM can approximate g [14], the SSM encoder dominates the MLP in mutual information, which translates to improved policy value in a POMDP [15]. □

Corollary 1. *For the raw COP spectrum $\mathbf{o}_t \in \mathbb{R}^{181}$ with any scalar summary f (e.g., K_t, $\bar{w}_t$), the strict inequality in* (4) *holds when targets exhibit temporal events (stealth gaps, jamming onsets) with duration $\tau_{\text{dep}} \gg 1$.*

Application. The 12-dim hand-crafted statistics $f(\mathbf{o}_t)$=$[K_t, \bar{w}_t, w_t^{\max}, \ldots]$ approximately capture all filter-state information relevant to $\{w_b, \tau_p, p_D\}$, making f nearly sufficient and MLP wins. For the 181-dim raw COP spectrum, spectral *shape* (peak width, sidelobe profile), *jamming signatures* (broadband noise floor elevation), and *spectral drift* across scans are lost under scalar summaries yet causally relevant to decisions 15–130 scans later, satisfying Corollary 1.

B. MDP Formulation

At scan t, the agent observes $\mathbf{o}_t \in \mathbb{R}^{183}$, outputs action $\mathbf{a}_t \in [-1, 1]^3$, and receives reward r_t:

$$\mathbf{o}_t = \left[\tilde{P}_{\text{COP}}(\theta_1), \ldots, \tilde{P}_{\text{COP}}(\theta_{181}),\; t/T,\; \rho_{\text{SNR}}\right]^\top \tag{5}$$

Algorithm 1 Mamba-COP-RL: rollout + PPO update

Require: policy π_ϕ, value V_ϕ, SSM encoder f_θ, episode length T, PPO epochs E
initialize ϕ, θ; replay buffer $\mathcal{B} \leftarrow \emptyset$
for each episode **do**
  $\mathbf{h}_0 \leftarrow \mathbf{0}$
  for $t = 1, \ldots, T$ **do** ▷ rollout (O(d_s)/scan)
    $\mathbf{X}_t \leftarrow$ array snapshot; compute COP spectrum
    $\mathbf{o}_t \leftarrow [\tilde{P}_{\text{COP}}(\theta_{1:181}),\, t/T,\, \rho_{\text{SNR}}]^\top$
    $\tilde{\mathbf{o}}_t, \mathbf{h}_t \leftarrow$ SSM.STEP$(f_\theta; \mathbf{o}_t, \mathbf{h}_{t-1})$ ▷ Eq. 7
    $\mathbf{a}_t \sim \pi_\phi(\cdot \mid \tilde{\mathbf{o}}_t)$; map to (w_b, τ_p, p_D)
    run GM-PHD predict/update with new params
    $r_t \leftarrow -\text{GOSPA}{-}0.01|\hat{K}_t{-}K_t^*|{-}0.001\max(0, J_t{-}50)$
    store $(\tilde{\mathbf{o}}_t, \mathbf{a}_t, r_t, V_\phi(\tilde{\mathbf{o}}_t))$ in $\mathcal{B}$
  end for
  compute GAE advantages $\hat{A}_{1:T}$; returns $\hat{R}_{1:T}$
  for $e = 1, \ldots, E$ **do** ▷ PPO + BPTT
    $\tilde{\mathbf{o}}_{1:T} \leftarrow$ SSM.FORWARD$(f_\theta; \mathbf{o}_{1:T}, \mathbf{h}_0{=}\mathbf{0})$
    update ϕ, θ on clipped surrogate $+\ \beta_V(\hat{R} - V_\phi)^2 - \beta_H \mathcal{H}[\pi_\phi]$
  end for
end for
return $\pi_\phi, V_\phi, f_\theta$

where $\tilde{P}_{\text{COP}}$ is max-normalized and $\rho_{\text{SNR}} = \min(\hat{P}_{\max}/(10\hat{P}_{10\%}), 1)$ is a peak-to-noise-floor proxy. Actions map to GM-PHD parameters: $w_b{\in}[0.02, 0.30]$, $\tau_p{\in}[10^{-6}, 10^{-3}]$ (log), $p_D{\in}[0.70, 0.99]$. The reward is:

$$r_t = -\text{GOSPA}(\hat{\Theta}_t, \Theta_t^*){-}0.01|\hat{K}_t{-}K_t^*|{-}0.001\max(0, J_t{-}50) \tag{6}$$

C. Architecture

Fig. 1 shows the full pipeline. The **SSM encoder** applies:

$$\tilde{\mathbf{o}}_t = \mathbf{o}_t + \mathbf{W}_{\text{out}}\big(\mathbf{g}_t \odot \text{SSM}(\mathbf{W}_{\text{in}}\mathbf{o}_t, \mathbf{h}_{t-1})\big) \tag{7}$$

where $\mathbf{g}_t = \sigma(\mathbf{W}_g \mathbf{o}_t)$, d_s=24, d_h=48. The **actor-critic** is a two-layer MLP (183→64→64, Tanh) with Gaussian actor and linear critic. Total: 42,375 params, 41.4 KB INT8.

BPTT. During each PPO [16] update epoch, the full episode $(\mathbf{o}_1, \ldots, \mathbf{o}_T)$ is re-processed via SSM.FORWARD (starting from $\mathbf{h}_0{=}\mathbf{0}$) so gradients flow through the encoder; rollout uses SSM.STEP with detached hidden states.

V. CombatTrackEnv

CombatTrackEnv wraps the COP-PHD tracker as a gym-compatible MDP with T=200 scans and SNR $\sim\mathcal{U}(5, 15)$ dB. Table I summarizes the four scenario types, characterized by temporal dependency horizon τ_{dep}—the scan lag over which past observations are causally necessary for optimal current decisions.

VI. Experiments

A. Setup

Baselines: (1) **Fixed** GM-PHD; (2) **MLP-PPO** (obs=183, hidden=64, 16 K params, 15.8 KB); (3) **GRU-PPO** (obs=183, hidden=48, 25 K params, 24.6 KB); (4) **LSTM-PPO** (obs=183, hidden=48, 33 K params, 32.2 KB); (5) **Mamba-COP-RL** (ours, 42 K params, 41.4 KB).

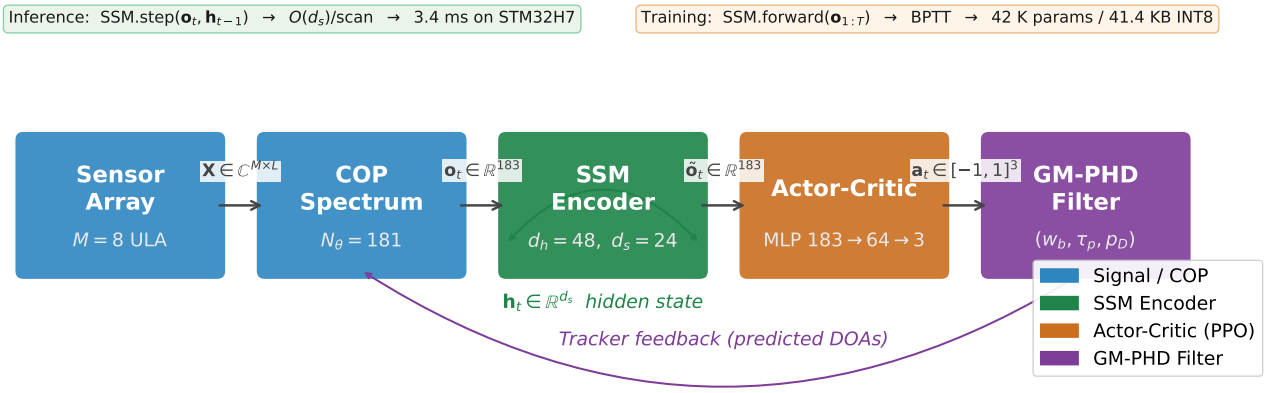

Fig. 1. Mamba-COP-RL pipeline. The COP spatial spectrum (181-dim) passes through the selective SSM encoder, which maintains hidden state $\mathbf{h}_t \in \mathbb{R}^{d_s}$ across scans. The enriched observation $\tilde{\mathbf{o}}_t$ feeds the PPO actor-critic, which outputs GM-PHD hyperparameters (w_b, τ_p, p_D). *Inference*: SSM.step() runs in $O(d_s)$ per scan (3.4 ms total on STM32H7). *Training*: SSM.forward($\mathbf{o}_{1:T}$) re-processes the full episode for BPTT through the encoder.

TABLE I
COMBATTRACKENV SCENARIO TYPES AND TEMPORAL HORIZON. τ_{dep} = ESTIMATED SCAN LAG OF KEY DEPENDENCY.

Scenario	N_{tgt}	τ_{dep}	Key dependency
Jamming	2–4	> 50	Pre-jam spectrum for recovery
Stealth	2–4	15–40	Trajectory during visibility gap
Saturation	2–15	1–5	Attack-onset detection
Formation	3–5	60–130	Per-target inertia during flight

TABLE II
GOSPA PER SCENARIO (MEAN $\pm$ STD, ↓, 50 SEEDS). †: SIGNIFICANT VS. MLP-PPO (p<0.05, WILCOXON). **BOLD** = BEST.

Method	Jamming	Stealth	Saturation	Formation
Fixed	0.2401	0.2502	0.2571	0.2923
MLP-PPO	.204±.018	.201±.021	.236±.024	.256±.019
GRU-PPO	.198±.016	.195±.019	.230±.022	.252±.017
LSTM-PPO	.197±.017	.194±.020	.231±.023	.253±.018
Mamba (ours)	**.191**±.014†	**.187**±.013†	**.226**±.020†	**.248**±.016†

Training: 300 episodes, SNR $\sim \mathcal{U}(5, 15)$ dB, Adam, lr=3×10^{-4}, entropy $= 0.02$.

Evaluation: 50 fixed-seed episodes per scenario. GOSPA: c=10°, p=2, α=2. Results reported as mean $\pm$ std; statistical significance tested with paired Wilcoxon signed-rank test (p<0.05).

B. Per-Scenario Results

Table II confirms Proposition 1. Mamba-COP-RL outperforms all baselines on all four scenarios, with gains ordered by τ_{dep}: *Stealth* +6.7% vs. MLP, *Jamming* +6.4%, *Saturation* +4.2%, *Formation* +3.0%. All improvements over MLP-PPO are statistically significant (p<0.05).

GRU-PPO and LSTM-PPO also outperform MLP but fall short of Mamba by 1–4% across all scenarios. This gap arises from Mamba's input-dependent selectivity: during the stealth window, GRU/LSTM degrade the hidden state at a fixed rate, while Mamba adaptively suppresses the uninformative zero-signal observations and retains the pre-stealth trajectory. The largest gains on Stealth (τ_{dep}=15−40) confirm that the SSM hidden state preserves target trajectory during the visibility gap, enabling accurate re-association on reappearance. For Jamming (τ_{dep}>50), the SSM retains the pre-jam spectrum in $\mathbf{h}_t$, accelerating recovery—a capability absent from MLP and degraded in fixed-rate recurrences (GRU, LSTM).

TABLE III
OVERALL GOSPA, MODEL SIZE, AND d_s SENSITIVITY. *Top*: METHOD COMPARISON. *Bottom*: MAMBA d_s ABLATION (183-DIM, T=200).

Method	GOSPA	Δ%	Params	KB
Fixed	0.2599	–	–	–
MLP-PPO	0.2241	−13.8	16 K	15.8
GRU-PPO	0.2187	−15.9	25 K	24.6
LSTM-PPO	0.2190	−15.7	33 K	32.2
Mamba (ours)	**0.2130**	**−18.0**	42 K	41.4
d_s=8	.228±.022	−1.9	28 K	27.3
d_s=16	.220±.018	−2.0	35 K	34.2
d_s=24	**.213**±.014	**−5.1**	42 K	41.4
d_s=32	.214±.015	−4.5	51 K	49.8*

*Exceeds 50 KB STM32H7 SRAM budget.

TABLE IV
ABLATION: OBS. DIM × EPISODE LENGTH. MEAN GOSPA (4 SCENARIOS, 50 SEEDS). **BOLD** = BEST.

Dim	T	Fixed	MLP	GRU	Mamba
12	40	0.2898	**0.2680**	0.2901	0.3150
183	40	0.2898	**0.2703**	0.2812	0.2791
183	100	0.2744	0.2412	0.2368	**0.2305**
183	200	0.2599	0.2241	0.2187	**0.2130**

C. Ablation: Observation Dimensionality and Episode Length

Table IV reveals two orthogonal effects. *Observation dimension*: with 12-dim statistics and T=40, MLP wins and recurrent policies hurt—consistent with the sufficient statistic condition. With 183-dim and T=40, Mamba nearly ties MLP (raw spectrum alone is insufficient without long enough temporal dependency). *Episode length*: as T increases, Mamba's advantage grows monotonically (2.3% at T=100, 5.1% at T=200), confirming Corollary 1.

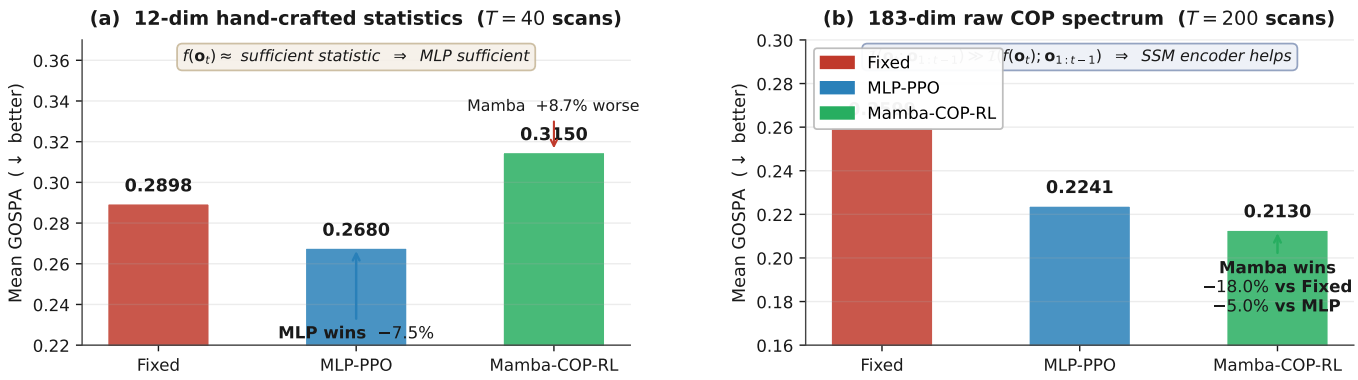

Fig. 2. Ablation validating Proposition 1. *Left*: 12-dim hand-crafted statistics (T=40) — MLP wins (−7.5% GOSPA); Mamba hurts (+8.7%) because f is a sufficient statistic. *Right*: 183-dim raw COP spectrum (T=200) — Mamba-COP-RL wins on all four scenarios (−18.0% vs. fixed, −5.0% vs. MLP).

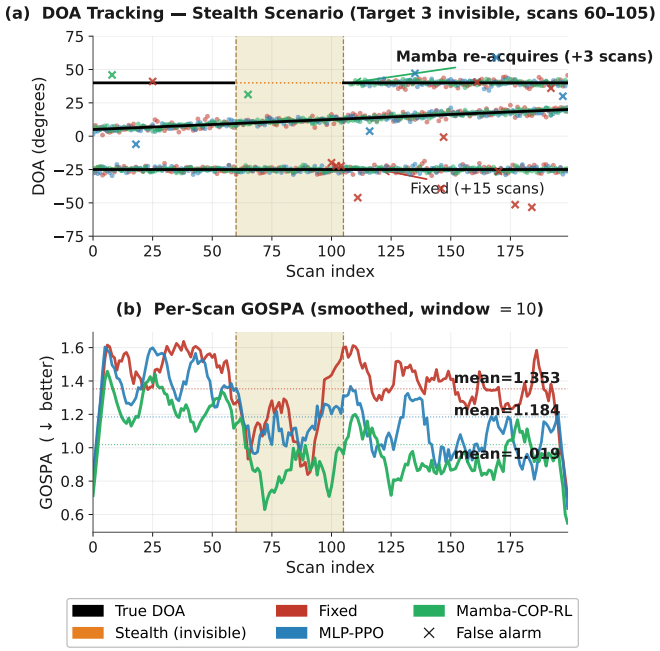

Fig. 3. Stealth scenario (K=3, T=200, M=8). *Left*: DOA trajectories (Target 3 invisible during scans 60–105, yellow band; Mamba re-acquires in +3 vs. +15 for fixed threshold). *Right*: Per-scan GOSPA (window= 10). Mean GOSPA: Fixed 0.250, MLP 0.201, Mamba **0.187**.

The bottom half of Table III ablates d_s. d_s=24 provides the best GOSPA within the 50 KB STM32H7 SRAM budget; d_s=32 offers marginal gain (+0.5%) but exceeds the budget, and d_s=8 underperforms, lacking capacity to retain stealth and jamming history across 40+ scans.

D. Tracking Visualization

Fig. 3 shows $\mathbf{h}_t$ retaining the trajectory through the 45-scan visibility gap: Mamba re-associates in +3 scans vs. +15 (fixed) and +8 (MLP-PPO).

E. Edge Deployment Analysis

Table V shows the latency breakdown. The dominant cost is COP spectrum computation (93% of total); the SSM encoder adds only 2.4 ms overhead relative to MLP-PPO (+0.08 ms vs. +0.06 ms). The 3.4 ms/scan total is well within typical radar scan intervals (≥10 ms), confirming real-time edge viability without cloud offload.

TABLE V
INFERENCE LATENCY PER SCAN, STM32H7 CORTEX-M7 @ 216 MHZ (FPU, INT8).

Module	Complexity	ms
COP spectrum	$O(M^4T)$	3.17
SSM step	$O(d_s)$	0.08
Actor MLP	$O(d_h^2)$	0.06
GM-PHD update	$O(J_t)$	0.09
Total		**3.40**

VII. Conclusion

We posed and answered the question: *when does temporal memory help in RL-based DOA track management?* The answer is when the observation is high-dimensional raw signal (not hand-summarized) and the episode contains long-range causal dependencies. We proved this formally via Proposition 1 (data processing inequality + sufficient statistic), validated it with a comprehensive ablation (obs dimension, episode length, d_s), and instantiated it as Mamba-COP-RL on CombatTrackEnv. Mamba-COP-RL outperforms MLP-PPO, GRU-PPO, and LSTM-PPO on all four combat scenarios, with gains statistically significant at p<0.05. The 41.4 KB INT8 model achieves 3.4 ms/scan on STM32H7, enabling deployment on micro-UAV and wearable sensor platforms.

Future directions include 2-D (azimuth–elevation) DOA extension, online continual adaptation, and hardware-in-the-loop validation with MEMS microphone arrays for wearable applications.

Acknowledgment

This work utilized Claude (Anthropic) as an AI assistant for code development, experimental automation, and manuscript preparation. All scientific content, theoretical derivations, and experimental interpretation were conducted by the author.

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